

BMS_SLAVE Fabrication Document

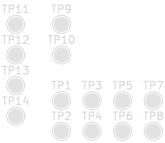
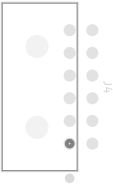
Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen				Legend	GBR
F,Mask		0,02mm		Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0,035mm (1,00oz)		Signal	GBR
Core		1,209994mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L4 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
B,Mask		0,02mm		Solder Mask	GBR
B,Silkscreen				Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,699994mm
Note: external layer thicknesses are specified after plating

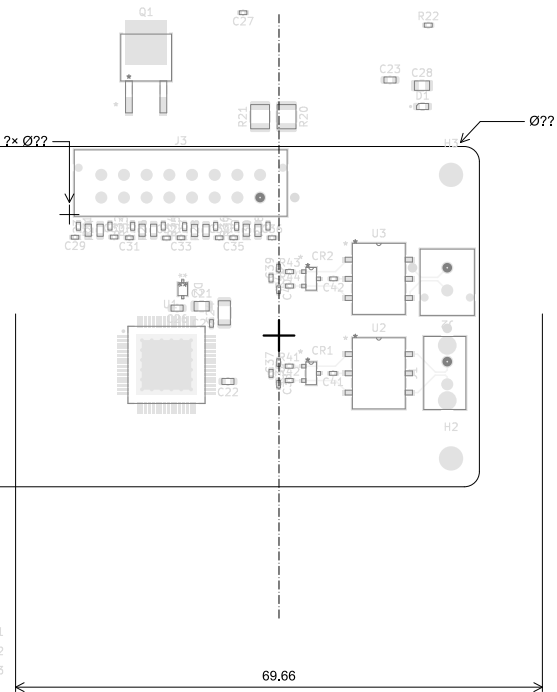
Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2



FID1
FID2
FID3

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BLUE DEVIL MOTOR SPORTS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE75.000 × 45.000 mm

BOARD THICKNESS1.660 mm

TRACE WIDTH0.200 mm

TRACE TO TRACE-0.000 mm

MIN. HOLE (PTH)0.200 mm

MIN. HOLE (NPTH)1.050 mm

ANNULAR RING0.150 mm

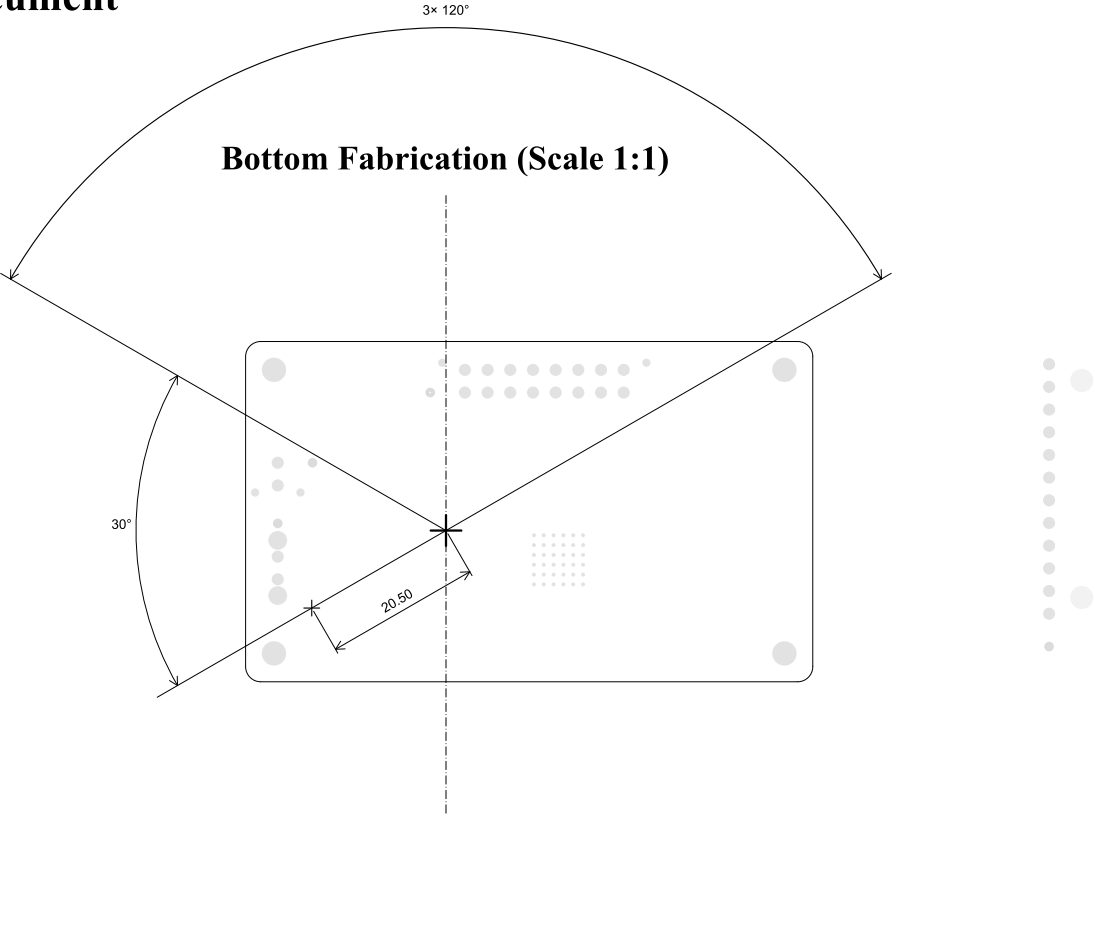
COPPER TO HOLE0.254 mm

COPPER TO EDGE0.250 mm

HOLE TO HOLE0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

	Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 2fd0761
	Sheet Title: Top Fabrication (Scale 1:1)	Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Path:	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
			Reviewer:	Size: A4	Sheet: 1 of 10

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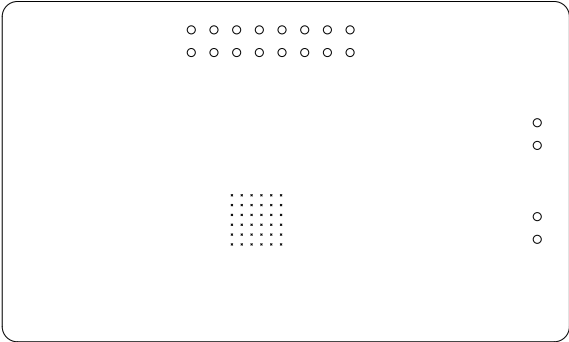
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			Blue Devil Motor Sports		PRELIMINARY		2fd0761		
			Board Name:			Project Name:			
			BMS_SLAVE			BMS Slave			
Sheet Title:		File Name:		Designer:		Date:		Revision:	
Bottom Fabrication (Scale 1:1)		bms_slave.kicad_pcb		Aidan Brzezinski		2024-04-13		+ (Unreleased)	
Sheet Path:				Reviewer:		Size:		Sheet:	
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	36	0.20mm (8,00mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
O	44	1.07mm (42,00mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
	Total 80					

Drill Drawing L1 - L4 (Scale 1:1)



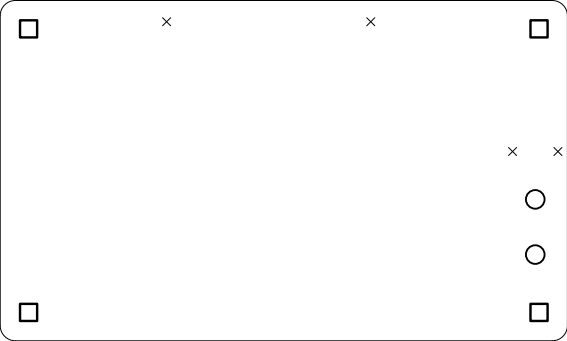
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	1.07mm (42.00mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
○	2	2.46mm (97.00mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
+	4	3.05mm (120.00mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
□	4	3.20mm (125.98mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
Total 14						

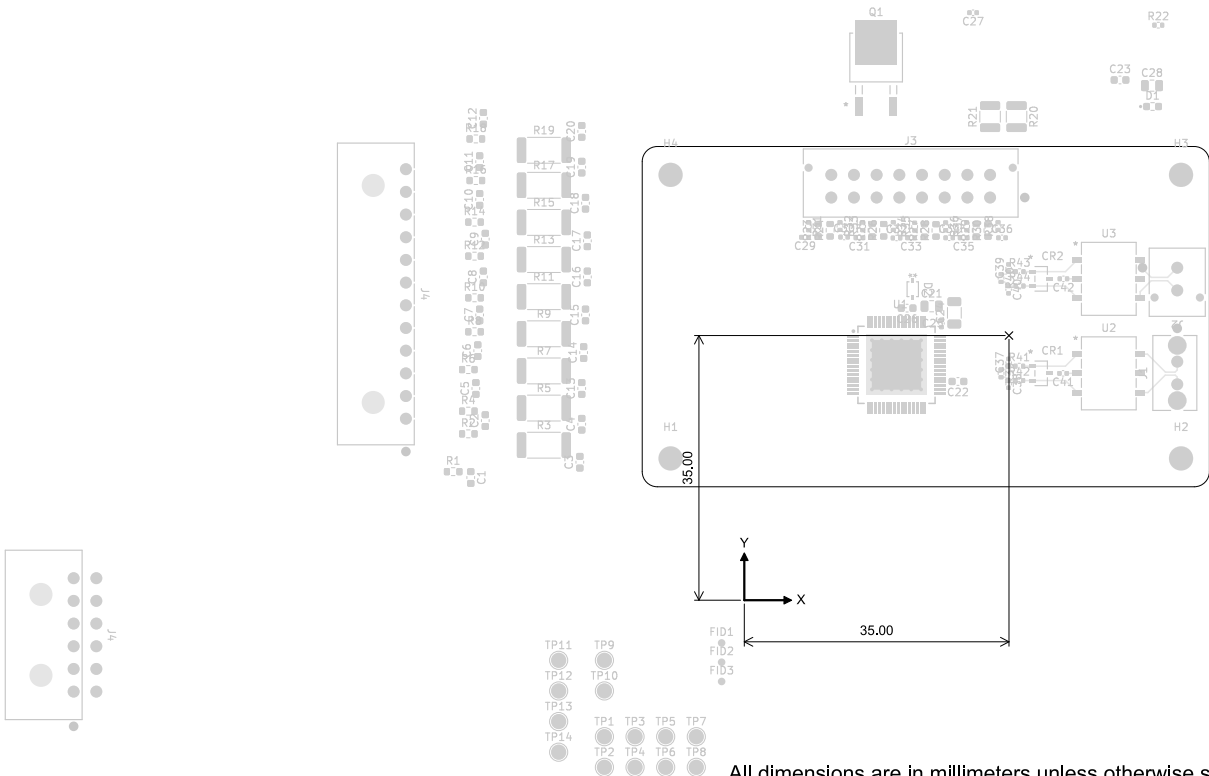
Drill Drawing L1 - L4 (Scale 1:1)



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Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP1	CVDD	-5.00	-33.05
TP2	DVDD	-5.00	-37.10
TP3	AVDD	-0.95	-33.05
TP4	LDOIN	-0.95	-37.10
TP5	VCOLL	3.10	-33.05
TP6	LDOIN	3.10	-37.10
TP7	TSREF	7.15	-33.05
TP8	GND	7.15	-37.10
TP9	T_P1	-5.00	-22.95
TP10	GND	-5.00	-27.00
TP11	U1_COMLP	-11.05	-22.95
TP12	U1_COMLN	-11.05	-27.00
TP13	U1_COMHP	-11.05	-31.05
TP14	U1_COMHN	-11.05	-35.10

Ref.	Net	X [mm]	Y [mm]
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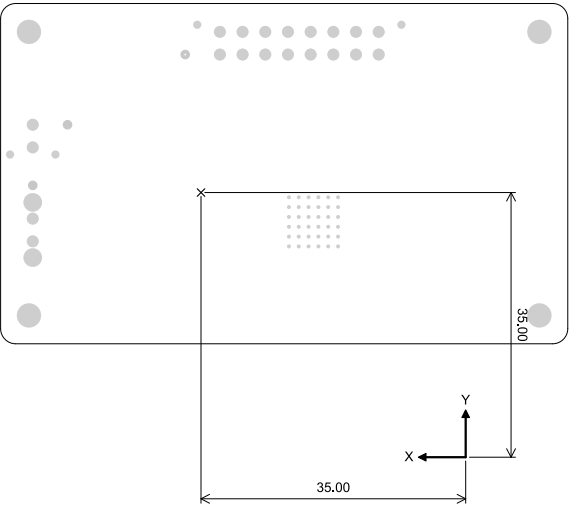
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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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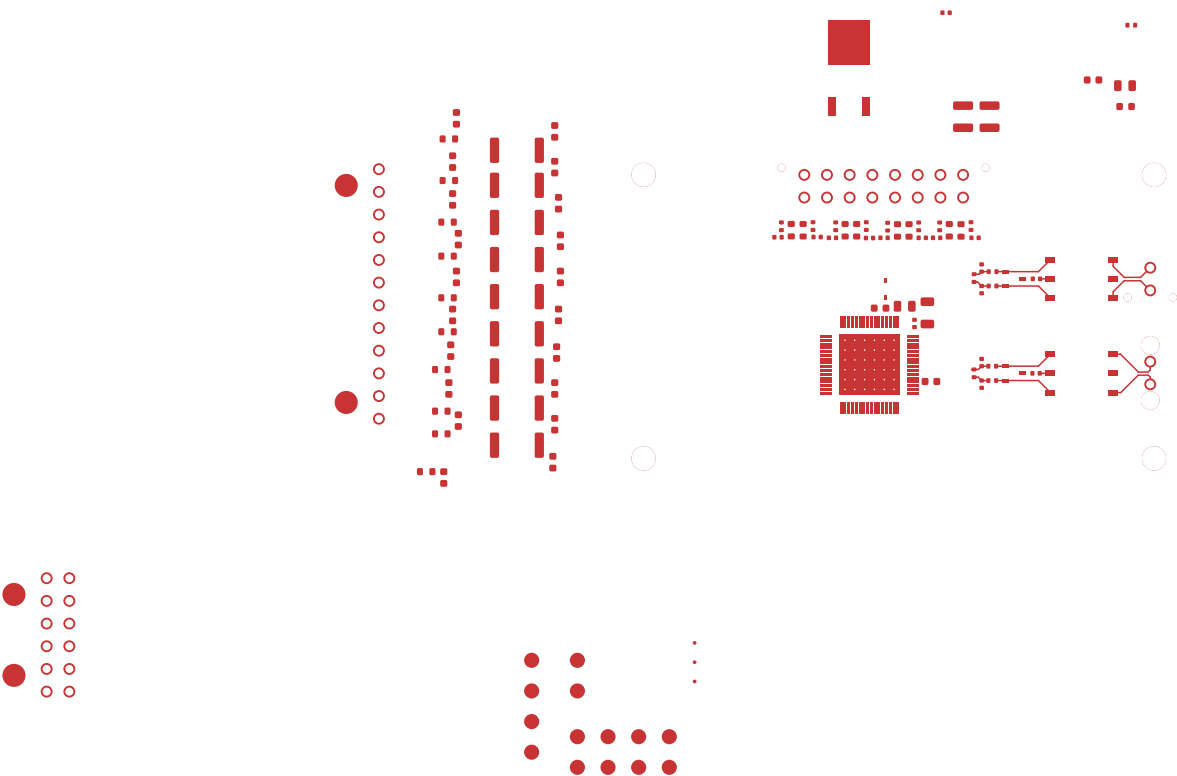


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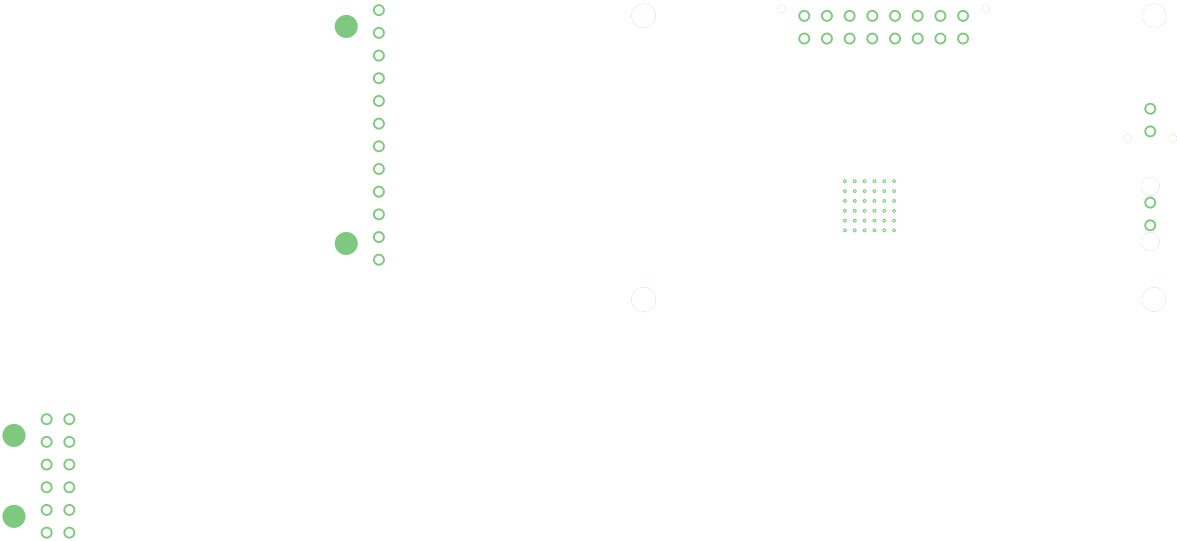
L1 (Sig, PWR) (Scale 1:1)



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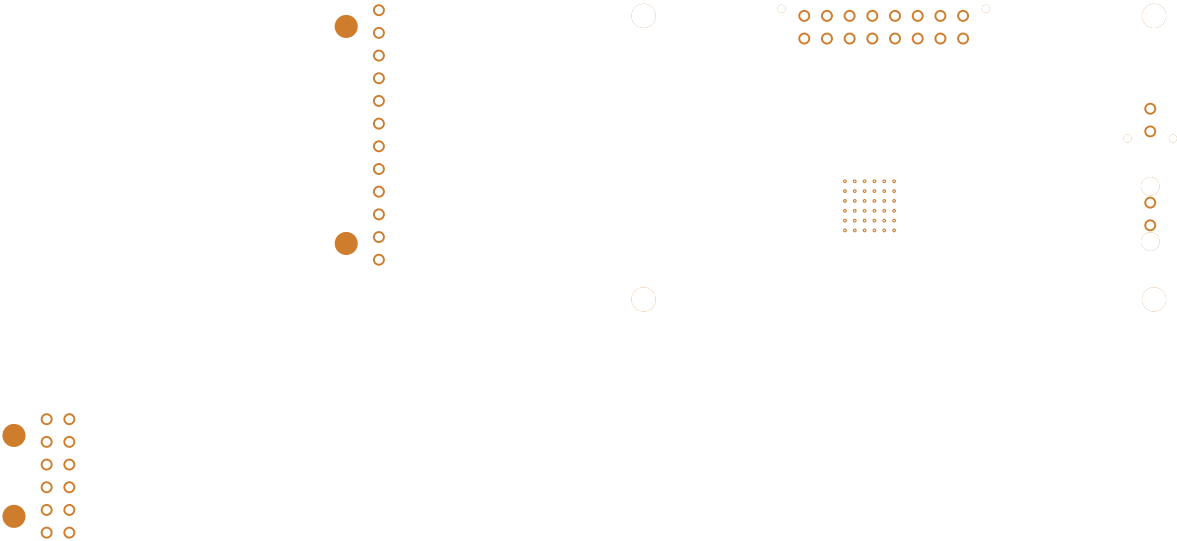
L2 (GND) (Scale 1:1)



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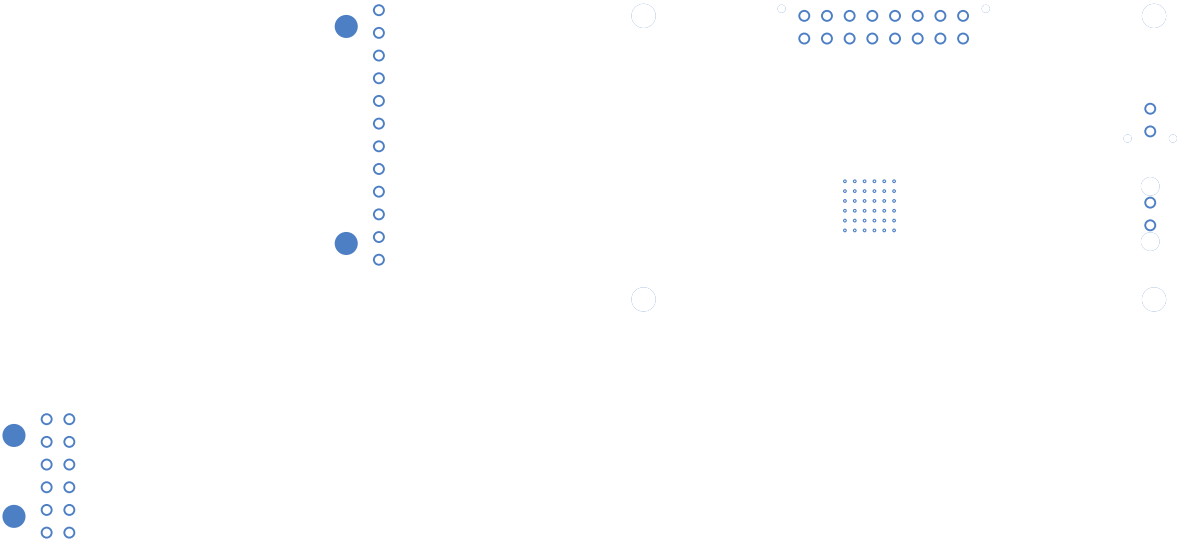
L3 (Sig, PWR) (Scale 1:1)



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L4 (Sig, PWR) (Scale 1:1)



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